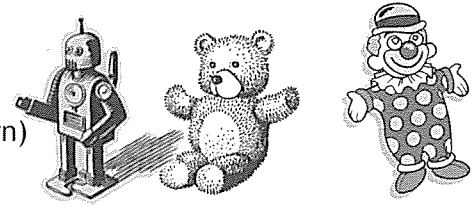


• PIE CHART

In a personality profile exercise, 120 fifteen-year-olds were asked to name the one toy (robot, teddy or clown) that they would most like to take home.

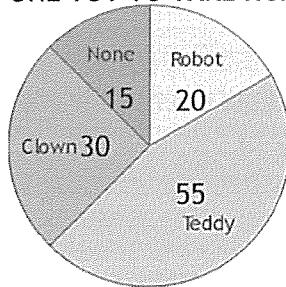


Robot

Teddy

Clown

ONE TOY TO TAKE HOME



Robot:	20 ≈ 16.7%
Teddy:	55 ≈ 45.8%
Clown:	30 = 25%
None:	15 = 12.5%
	<u>120 = 100%</u>

In the pie chart, each category is represented by a sector (a “slice of pie”).

$$\text{Angle at Centre} = \frac{\text{Number in Category}}{\text{Total Number of Observations}} \times 360^\circ$$

$$\text{For "Robot", Angle} = \frac{20}{120} \times 360^\circ = 60^\circ$$

$$\text{For "Teddy", Angle} = \frac{55}{120} \times 360^\circ = 165^\circ$$

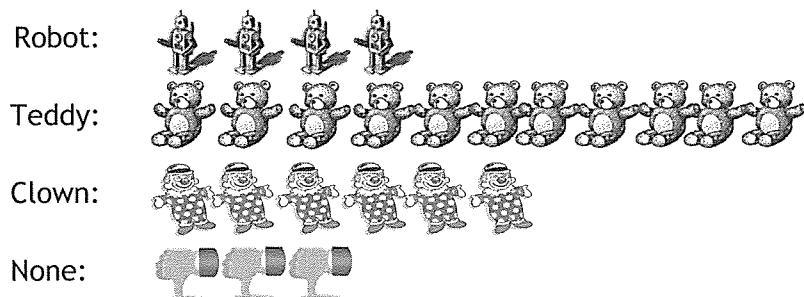
$$\text{For "Clown", Angle} = \frac{30}{120} \times 360^\circ = 90^\circ$$

$$\text{For "None", Angle} = \frac{15}{120} \times 360^\circ = 45^\circ$$

• PICTOGRAPH

The same information can also be represented in a pictograph, a series of small pictures or symbols:

TOYS THAT 15-YEAR OLDS CHOSE TO TAKE HOME



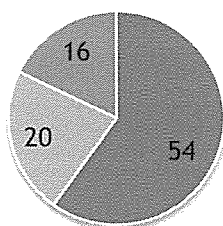
Key: Each symbol represents 5 fifteen-year-olds.

Exercise 5.2

Pie Charts & Pictographs

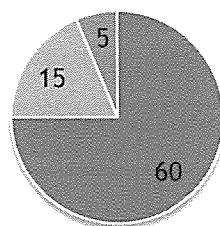
1. In a survey, supermarket customers were asked if they wanted earlier opening on Sundays. Their responses are shown in the pie charts:

Responses (Supermarket A)



- Yes, 54
- No, 20
- Undecided, 16

Responses (Supermarket B)



- Yes, 60
- No, 15
- Undecided, 5

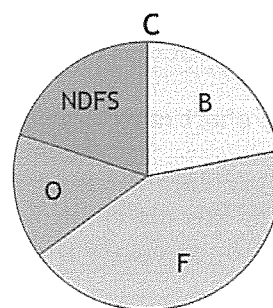
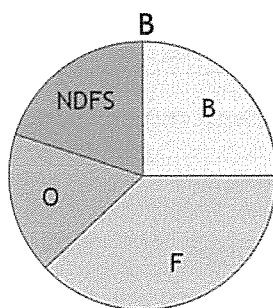
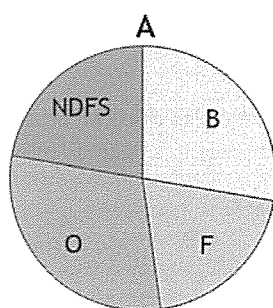
- a) How many customers were interviewed at each supermarket?
 b) Find the size of the three angles at the centre of each pie chart.
2. The main ingredients of “EnergiGrist” muesli are cereals. The rest is made up of nuts, dried fruit and sugar (NDFS).

In a 1 kg box of EnergiGrist, there are

- 250 g of Bran Logs (B),
- 380 g of Wheat Flakes (F)
- 170 g of Oat Clusters (O)
- 200 g of Nuts, Dried Fruit and Sugar (NDFS)



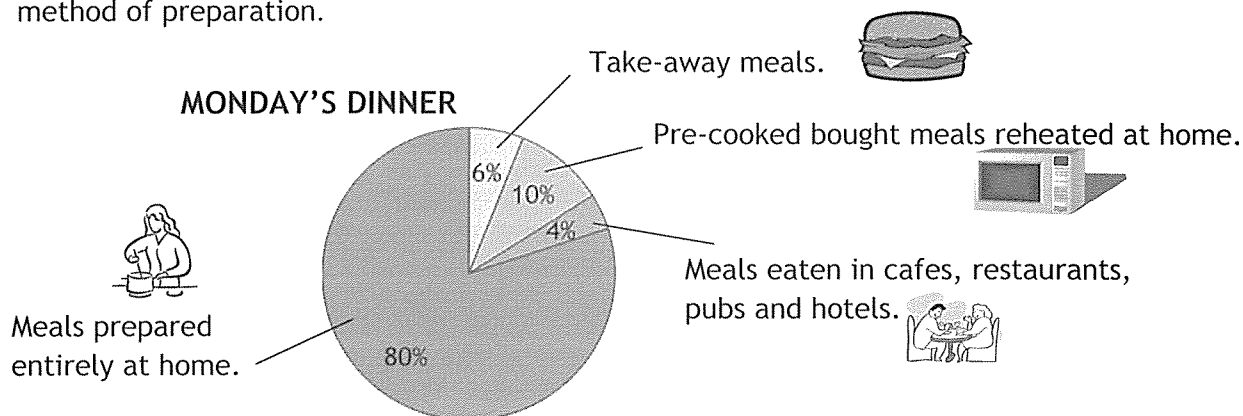
Only one of the pie charts (below) correctly illustrates the contents of EnergiGrist.



By calculating the size of the four angles at the centre of the pie chart, determine which chart is correct. Explain briefly.

Exercise 5.2 continued

3. The main meal of the day in Australian households may be classified according to its method of preparation.



- a) What proportion of meals were not prepared entirely at home?

On Fridays, there are different proportions:

• Prepared entirely at home	73%
• Take-away meals	17%
• Pre-cooked bought meals reheated at home	6%
• Meals eaten in cafes, restaurants etc.	4%

- b) What would be the angles at the centre of the Friday pie chart for each of the categories?

4. Some primary school children have professional after-school recreation activities. For example: ballet, tennis lessons, tae kwan do, horse riding, guitar lessons, cooking classes, swimming lessons, acting classes etc.

A random sample of 150 children from the Perth metropolitan region revealed:

No. of Days of Activities:	Zero	One	Two	Three	Four	Five	TOTAL
No. of Students:	52	42	32	17	2	5	150

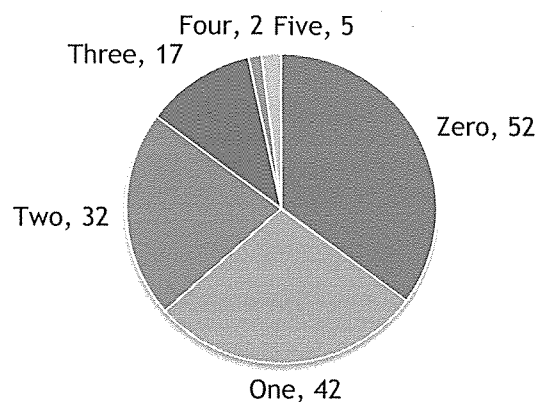
A pie chart may be used to illustrate this result:

No. of Days of Activities

- a) Calculate the largest central angle for this pie chart.
Also, calculate the smallest angle.

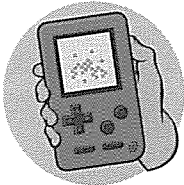
- b) What percentage of Perth Metropolitan region children have at least two days of professional after-school recreation activities?

- c) Would it be reasonable to conclude from this study that most Perth metropolitan region primary school children like professional after-school recreation activities? Explain briefly.



5. “Game-a-Lot” is a shop which rents out (or exchanges) games, DVDs, CDs and equipment.

For November, the shop’s income was gained from

	•Games	\$12 500	
	•DVDs	\$6300	
	•CDs	\$280	
	•Equipment	\$4120	

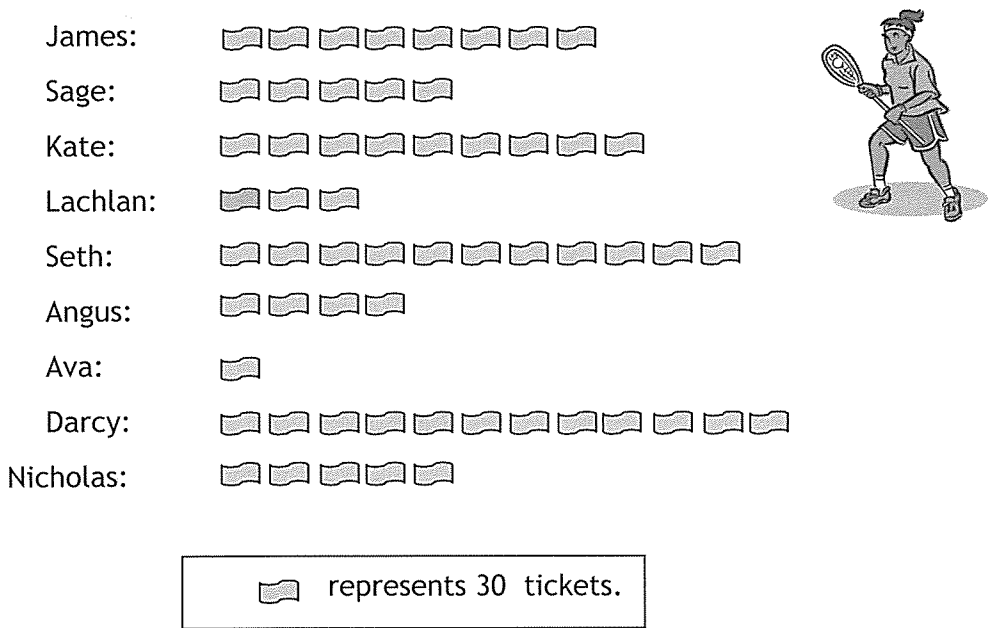
a) Calculate, to the nearest 1%, the percentage of total income for each of the four categories.

A pie chart with four sectors was used to illustrate this information.

b) Calculate, to the nearest degree, the four angles at the centre of the pie chart.

c) Draw the pie chart showing percentages for each category and a suitable title.

6. The lacrosse club held a raffle to raise funds. The number of raffle tickets sold by nine club members is shown in the pictograph.



a) Who sold the most tickets? How many?

The tickets come in books of thirty and each ticket costs \$2.

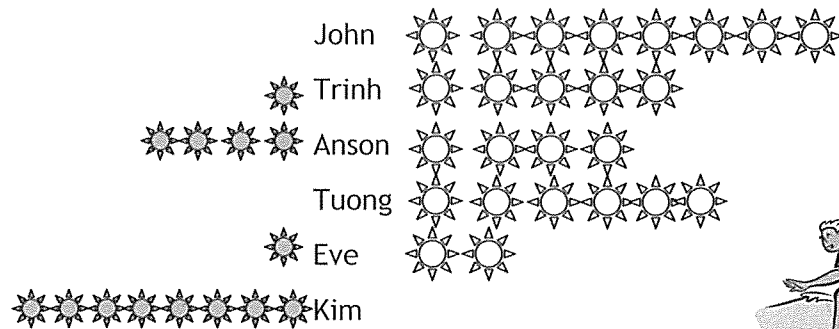
b) How much money was raised by Kate? by Seth?

c) How much money altogether was raised by the nine club members?

Exercise 5.2 continued

7. At the local surf-lifesaving club, junior members are expected to attend 12 hours of training sessions per month. Each day session lasts two hours. They are held on the weekend.

Some sessions are held during the week, in the evening. They last for $1\frac{1}{2}$ hours. The training record for the junior squad for January 2015 was



$\equiv$ One 2-hour day session
 $\equiv$ One $1\frac{1}{2}$ -hour evening session

- a) Complete the table (below) to show total training hours for these junior members:

	John	Trinh	Anson	Tuong	Eve	Kim
Day:	16				4	
Evening:	0	$1\frac{1}{2}$			$1\frac{1}{2}$	
TOTAL:		$11\frac{1}{2}$			$5\frac{1}{2}$	

- b) Did any junior members fail to meet their training obligations? Who?
- c) What percentage of the total training hours was evening training?
8. The Go-'n'-Gulp drive-through coffee shop keeps a monthly sales record for each of its counter staff: Scuzz, Ben and Fido.

The print-out shows sales for beverages and for snacks in October:



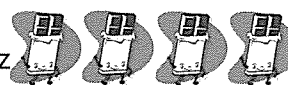
$\equiv$ 10 beverages



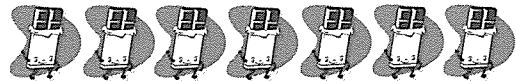
$\equiv$ 10 snacks



Scuzz



Ben



Fido



- a) What were the total sales for October for beverages and for snacks?

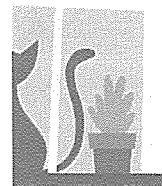
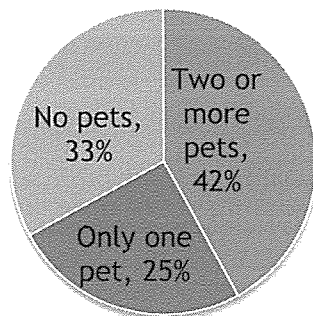
The profit from beverages is \$3.15 per cup and for snacks, \$2.90 per item.

- b) Which one of the counter staff delivered the most profit in October?

List the individual profits for Scuzz, Ben and Fido.

9. Winston Avenue contains 84 households. A survey has been conducted to count the number of household pets. The categorical data is recorded in the pie chart below.

Pets on Winston Avenue



How many households have

- a) no pets? b) one pet? c) two or more pets?
d) at least one pet? e) less than two pets?
f) Which is the best estimate of the total number of pets in Winston Avenue?

15 or 35 or 70 or 100

Explain why you chose that answer.

- g) Comment on these other sets of categories for classifying the number of pets in Winston Avenue?

Furry, Not Furry

Tall, Short

Feathers, Fur, Scales

Pink, Green, Yellow, Black

Small, Medium, Large

Cats, Dogs, Birds

No legs, Two legs, Four legs

Swims, Walks, Flies

